A counterexample to the core partition component formula

TLMC conjecture 00000003829

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Abstract

The coprime component formula in conjecture 00000003829 fails at $(s,t) = (2,3)$. There are exactly two simultaneous cores, the empty partition and the one-box partition. A single-box border joins them, so the graph has one connected component, whereas the asserted rational Catalan value is two. The complete classification, the border, and the component-count contradiction are verified in a self-contained Lean 4 project using ordinary kernel-checked proofs.

1 Statement and border convention

The conjecture defines $B_{s,t}$ as the graph of (s,t)-core partitions generated by adding and removing borders, and asserts for coprime s, t that

$$\#\pi_0(B_{s,t}) = \frac{1}{s+t}\binom{s+t}{s}.$$

A partition is a finite weakly decreasing sequence of positive integers; the empty partition is included. Its diagram has λ_i cells in row i. The hook length is the number of cells strictly right of a cell, plus those strictly below it, plus one. A t-core has no hook length divisible by t.

We use the usual border-strip (rim-hook or ribbon) convention: a nonempty skew diagram connecting two partitions, connected through shared cell edges and containing no 2×2 square. In particular, a single cell is a border strip. These standard definitions appear in Jouhet and Wahiche, page 2 [2]. The same counterexample applies to the single-box Young graph, and to any border relation that retains the single-box edge used below.

The filed English and Chinese statements impose no restriction on border length. If a different graph is intended, for example one allowing only borders of lengths s or t, its edges need to be specified separately; the present counterexample does not assert the same conclusion for that modified definition. The Lean project explicitly records its graph rather than leaving the adjacency convention implicit.

2 Classification without a size bound

Lemma 1. *The simultaneous* $(2,3)$*-core partitions are exactly* $\varnothing$ *and* (1).

Proof. Both displayed diagrams have no hook divisible by 2 or 3. For the converse, induct on the number of rows. Removing the first row does not change any hook in the lower rows, so the remaining partition is itself a simultaneous core. By induction it is empty or (1).

If it is empty, the original partition is one row of length $x > 0$. If $x \geq 2$, its second cell from the right has hook length 2, a contradiction. Thus $x = 1$.

If the remaining partition is (1), the original partition has row lengths $(x, 1)$. When $x = 1$, the top-left cell has hook length 2. When $x = 2$, it has hook length 3. When $x \geq 3$, the second cell from the right in the first row has no cell below it and has hook length 2. Every possibility contradicts a core condition. Thus this case cannot occur, completing the induction. □

Theorem 1. *Under the stated border convention, conjecture 00000003829 is false.*

Proof. Take $s = 2$ and $t = 3$. The graph has precisely the two vertices from the lemma. Adding the cell $(0, 0)$ to the empty diagram produces (1). The skew diagram consists of this one cell, so it is nonempty, edge-connected, and contains no 2×2 square. Hence the two vertices are adjacent and their component quotient has cardinality one. On the other hand,

$$\gcd(2, 3) = 1, \qquad \frac{1}{2+3}\binom{2+3}{2} = \frac{10}{5} = 2.$$

The component count is therefore different from the claimed value. The coprime clause already fails, irrespective of the conjecture's additional noncoprime clause. □

3 Formal verification and reproducibility

The file `Basic.lean` imports only `Std` and is pinned to Lean `4.31.0`. It does not impose an enumeration cutoff on partitions. Its main definitions and theorems are:

- `hook`, `IsPartition`, and `IsCore` encode ordinary diagrams and every hook in every row.
- `classify` proves the lemma for arbitrary finite lists of positive row lengths satisfying both core conditions. This is slightly stronger than needed, since the proof does not use row monotonicity.
- `cells`, `skew`, and `Border` encode actual Young diagrams and border strips; `singleton_border` verifies the required edge.
- `Vertex` contains all simultaneous cores. `Reach` is undirected path equivalence and `Component` is its quotient. `HasCard` asserts the existence of a bijection with $\mathrm{Fin}(n)$.
- The quotient has one element (`component_count_one`) and cannot have two (`component_count_not_two`).
- `connected_for_any_border_relation` proves that any edge relation on these vertices containing the singleton edge is connected. Thus additional allowed borders cannot change the result.
- `counterexample` combines coprimality, cardinality one, and the failure of the conjectured value. Pascal recursion calculates $\binom{5}{2} = 10$ and $10/5 = 2$ exactly; no rounding affects the witness.

With the pinned toolchain on the executable search path, run `./reproduce.sh`. It builds the project, re-elaborates the proof, and replays its declarations through `leanchecker`. The axiom audit for the final theorem contains only Lean's standard `propext` and `Quot.sound`. No unproved assumptions, external computation oracle, or kernel bypass is used. The supplied `build.log` records the successful run.

References

[1] The Last Math Competition. *Conjecture 00000003829*, retrieved 11 October 2026. Official statement at the checked upstream commit.

[2] Frédéric Jouhet and David Wahiche. *Congruences for hook lengths of partitions*, page 2. `https://math.univ-lyon1.fr/~jouhet/congruences_pams_vf.pdf`.